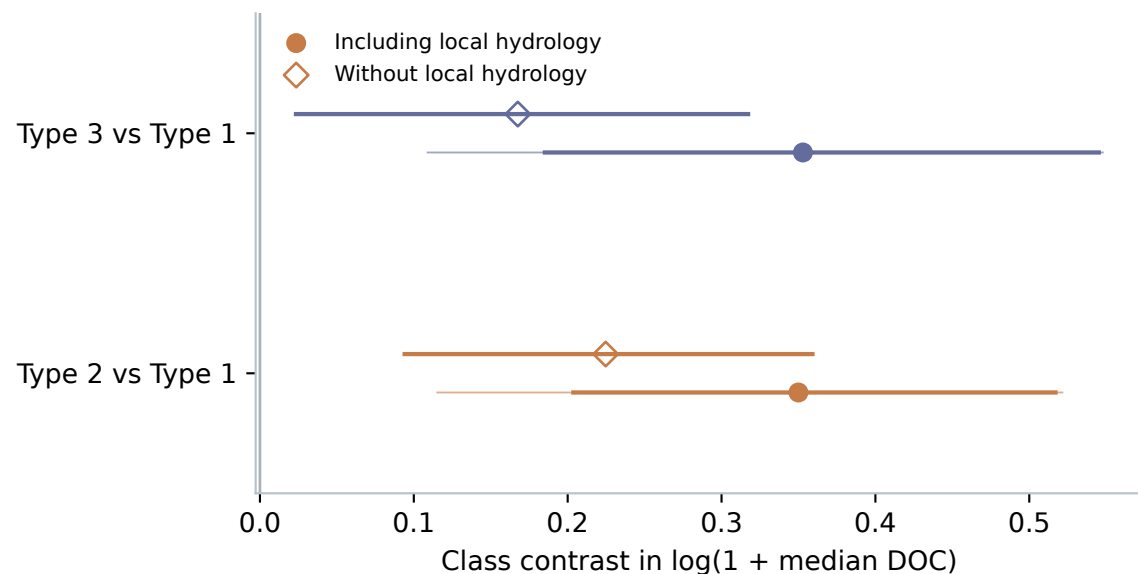
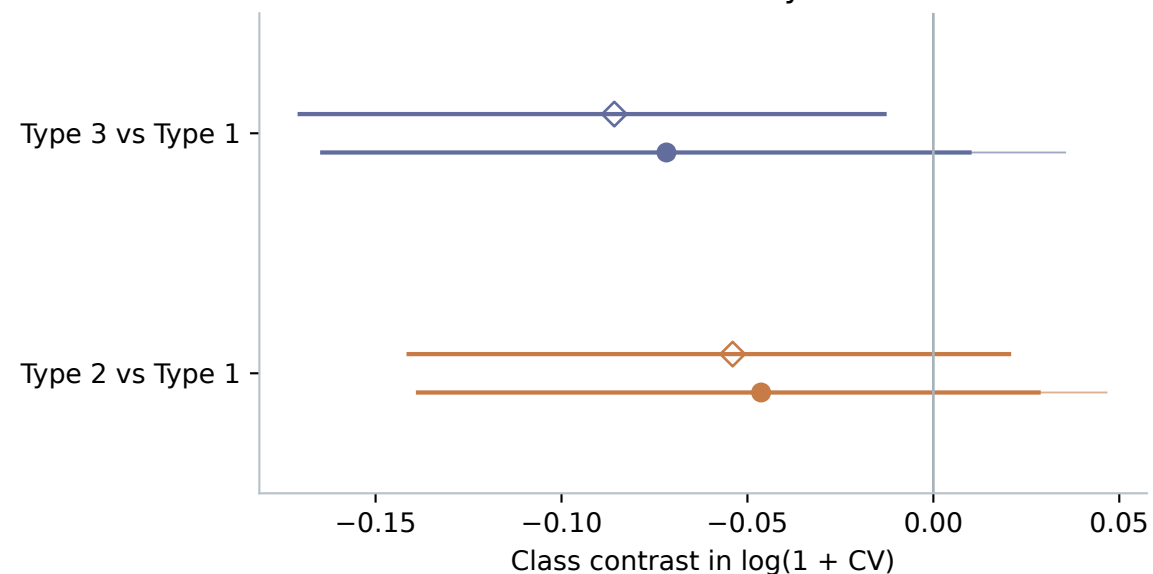


River structure associations after environmental adjustment

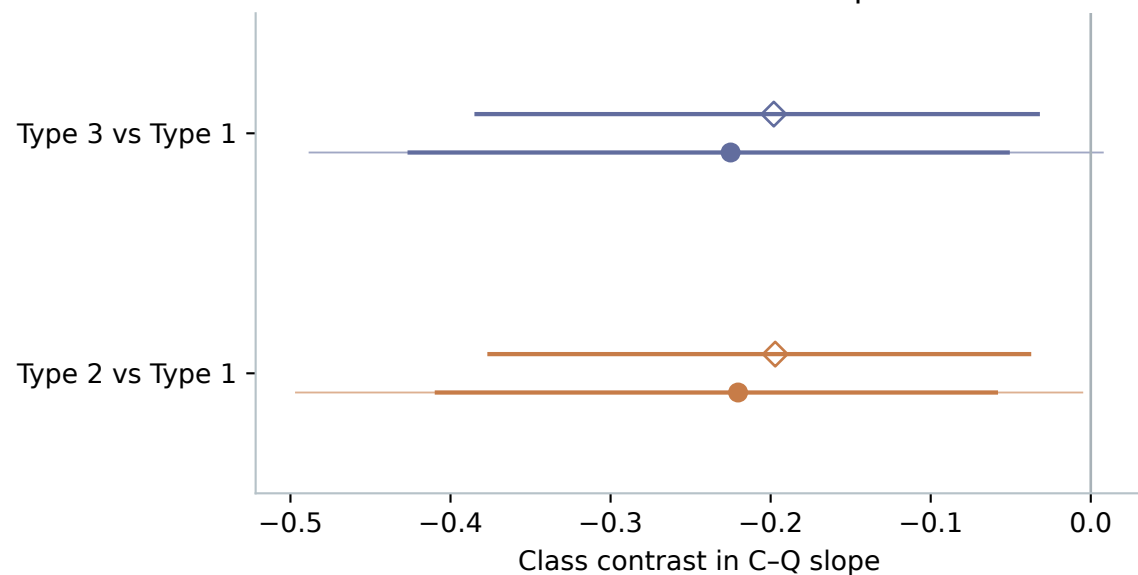
a Concentration



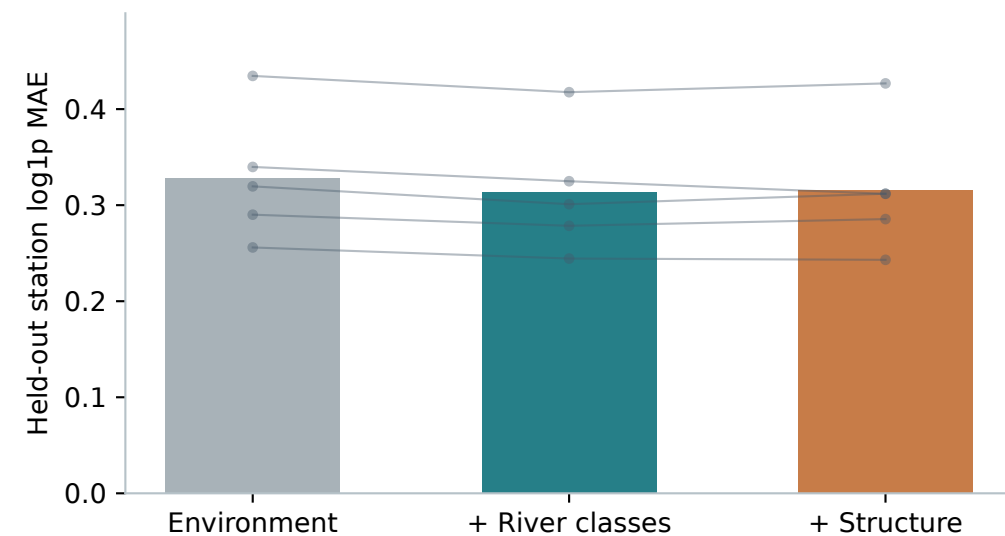
b Variability



c Concentration-flow response



d Five HUC4-blocked folds



Station-equal regression; controls include ecology, climate, sampling, geography and HUC2. Bootstrap intervals condition on the fixed classification. Thin intervals additionally resample HUC4 basins. Open diamonds omit local hydrology. Bars: equal-fold mean; connected dots: folds, ridge $\alpha=10$. Classes use all available physical geometry, without DOC; this is exploratory association analysis, not independent external validation.